Dengue Surge Warning for Singapore: Can Recent Surveillance and Weather Data Flag Elevated Risk?

Tran Duy Tai Do Nhat Thanh Nguyen Thi Quynh Thu

42.510 Statistical Learning

Abstract

Can recent dengue counts and weather identify weeks with a materially elevated risk of a dengue surge in Singapore? We forecast 1-, 2- and 4-week-ahead surges in weekly dengue fever (DF) notifications from the Ministry of Health bulletin (2012–2022, 574 weeks), with lagged rainfall and temperature from NEA/MSS Changi station and reanalysis humidity. Surges are defined in three ways (above a rolling 52-week mean + 2 SD, above the same-season 90th percentile, or a rise of at least 25% and 50 cases), plus onset-only versions that exclude weeks already above the threshold. Seasonal and persistence baselines are compared with logistic, lasso, CART, random forest, Poisson and ridge count models in weekly expanding-window rolling-origin forecasts (target weeks 2016–2021), with block-bootstrap intervals. 2022 is reported separately as a final-period check. For level surges, persistence already reaches PR-AUC 0.93–0.97 and no model beats it reliably. Count regression improves growth surges (PR-AUC 0.26 vs 0.17) and seasonal onsets (0.89 vs 0.26, but only 5 events). Weather adds no measurable skill beyond recent counts. These are predictive, not causal, results; few outbreaks, missing serotype data and prior exposure to 2022 limit them.

1 Problem and relevance

Dengue is endemic in Singapore, with substantial year-to-year variation and recurring epidemics. An early warning could help public-health teams prioritize vector control, risk communication and service readiness. Singapore forecasting studies have used climate and surveillance information at longer horizons (Hii et al., 2012; Shi et al., 2016); here we ask a narrower predictive question: *for Singapore dengue surveillance teams, can case counts and weather available by week t identify a materially elevated DF week at $t + h$?*

We compare three operationally distinct outcomes: crossing a recent-count threshold, crossing a historical seasonal threshold, and a relative increase with a minimum count. The onset variants ask the more decision-relevant question of whether an elevated period can be flagged before the series is already above its threshold. We then assess whether weather improves on recent cases and transparent seasonal or persistence baselines. The analysis is predictive, not causal: an association between weather and future counts does not identify the effect of weather or the impact of an intervention.

2 Data curation

The main outcome is weekly national DF cases from the Ministry of Health Weekly Infectious Disease Bulletin on data.gov.sg, 2012–2022 (574 epi weeks) (Ministry of Health, Singapore, 2024b). Dengue haemorrhagic fever (DHF) is retained separately but not modelled because it averages less than one case per week. We use NEA/Meteorological Service Singapore (MSS) daily rainfall and temperature at Changi station S24, aggregated to Monday–Sunday weeks. Weekly relative humidity comes from the Open-Meteo archive because MSS daily files do not include humidity (Meteorological Service Singapore; Open-Meteo). The brief's MOH 2014–2018 file is an independent source check and secondary analysis (Ministry of Health, Singapore, 2024a).

MOH epi weeks follow the MMWR Sunday–Saturday calendar (2014 and 2020 have week 53). Each week is keyed by the following Monday to align with weather. The bulletin and the 2014–2018 file agree exactly on 209 of 261 overlapping DF weeks; the maximum weekly difference is 8 cases and totals differ by about 0.2%. The largest DHF difference is one case. There are no duplicate or missing case weeks. Weather quality checks found no implausible values. MSS and Open-Meteo weekly temperature correlate at $r = 0.91$ (Open-Meteo is 1.3°C cooler); rainfall correlation is $r = 0.48$. Open-Meteo humidity correlates with NEA monthly Changi humidity at $r = 0.56$ and is 5.8 percentage points higher on average.

Cases are aggregated national notifications, not individual records; no personal information is used. Weather and cases are joined by `week_start`, with weather lags calculated before joining. Source links, licences, retrieval manifest, variable definitions, inclusion rules and further QA results are documented in `docs/DATA.md`. These data choices limit interpretation: one station may not represent local rainfall, and reported cases reflect testing and reporting as well as transmission.

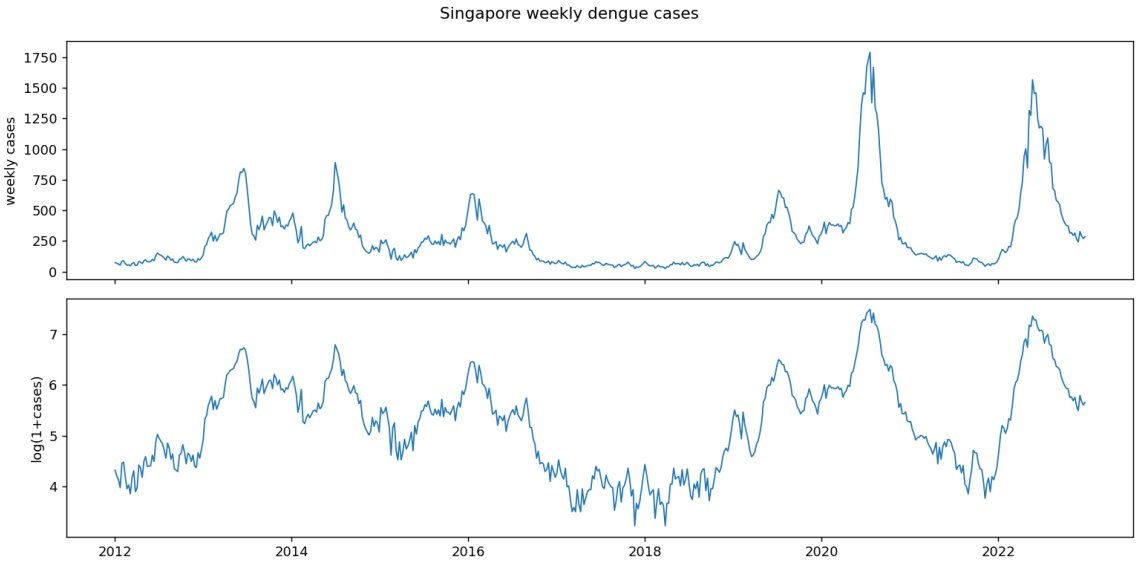


Figure 1: Weekly dengue fever counts in the main series; DHF is not included in the target.

3 Methods and validation

Let y_t denote weekly DF cases and $h \in \{1, 2, 4\}$ the forecast horizon. We label a surge if (A) $y_{t+h} > \mu_t + 2\sigma_t$, where μ_t, σ_t use the previous 52 weeks; (S) y_{t+h} exceeds the 90th percentile for dates within ±17 days of the same season in earlier years, excluding observations from the previous 26 weeks; or (G) $y_{t+h} \geq (1 + p_h)y_t$ and $y_{t+h} \geq 50$, with $p_h = 0.25, 0.30, 0.45$. A-onset and S-onset are defined only when y_t is not already above the corresponding threshold. Thresholds use information available at t.

Predictors include log-transformed current and lagged counts, recent growth and rolling means, seasonal terms, and rainfall, rain days, temperature and humidity lagged by 0–12 weeks. We compare prior, seasonal-rate and persistence baselines with logistic regression on cases, weather or both; L1-regularised logistic regression with forward time-series tuning; a depth-3 CART; random forest; Poisson regression; and ridge regression on log counts. Count forecasts are converted to probabilities of exceeding the target using a log-normal predictive distribution based on training residuals. Model comparisons use PR-AUC and Brier score; alarms at probability 0.20 illustrate a missed-surge:false-alarm cost ratio of 4:1, not a clinically validated threshold.

We use expanding-window rolling-origin forecasts by week from 2016, refitting every week and purging training rows whose complete label horizon would not yet be observed. The development period is defined by target weeks before 2022; target weeks in 2022 form the final-period evaluation, including forecasts issued in late 2021 for 2022. During this rolling evaluation, earlier 2022 labels can enter

training for later forecasts only after they would have become observable. We report PR-AUC, precision, recall, alarms per year and Brier score/calibration; count-model errors are also reported as MAE/RMSE in cases per week. Uncertainty uses 500 moving-block bootstrap draws with 26-week blocks. A one-week reporting-delay sensitivity re-runs the primary one-week targets with counts from week t unavailable.

Holdout qualification. Earlier exploratory runs pooled forecasts through 2022 before the final split was selected. In addition, the rolling evaluation updates using earlier observed outcomes within 2022. Thus this is a final-period prequential check, not a strictly untouched static test set or independent external validation.

4 Results

Exploration. Annual DF ranged from 2,759 (2017) to 34,959 (2020) cases, with large epidemics in 2013–2014, 2019–2020 and 2022 (Fig. 1). The three surge definitions pick different weeks: at $h = 1$ the Jaccard overlap is 0.31 (A vs S), 0.11 (A vs G) and 0.05 (S vs G). In the development period (target weeks 2016–2021, 312 weeks) the prevalence is 13.8% (A), 14.1% (S) and 11.5% (G), but only 7 (A-onset) and 5 (S-onset) weeks are surges that start from below the threshold.

Table 1: One-week-ahead results. PR-AUC with 95% block-bootstrap interval; precision (P) and recall (R) at the pre-specified alarm threshold $P \geq 0.20$. "Best" = highest development PR-AUC among non-baseline models; the same model is then scored on 2022. 2022 intervals are omitted because 52 weeks hold only two 26-week blocks.

	Development, target weeks 2016–2021					Target weeks 2022		
Target	Pos.	Persist.	Best model: PR-AUC [95% CI]	P	R	Pos.	Persist.	Best
A threshold	43	0.93	Poisson GLM: 0.93 [0.78, 0.98]	0.62	0.93	20	0.98	0.97
S seasonal	44	0.97	Ridge log-count: 0.98 [0.77, 1.00]	0.75	1.00	27	0.98	0.97
G growth	36	0.17	Ridge log-count: 0.26 [0.15, 0.40]	0.22	0.53	8	0.14	0.31
A-onset	7	0.30	Poisson GLM: 0.32 [0.04, 0.80]	0.17	0.57	2	0.42	0.23
S-onset	5	0.26	Ridge log-count: 0.89 [0.70, 1.00]	0.33	1.00	2	0.58	1.00

Development comparison (Table 1, Fig. 2). For the level targets A and S, the one-variable persistence baseline (current distance to the threshold) already reaches PR-AUC 0.93–0.97, and no model beats it reliably: the share of bootstrap draws in which the best model beats persistence is 0.37 (A) and 0.70 (S). These high scores mostly mean that a week above the threshold is followed by another one. At the 0.20 threshold the A model raises about 11 alarms a year and catches 93% of surge weeks. Growth surges are hard: the best model (PR-AUC 0.26) is better than persistence (0.17) in 98% of bootstrap draws, but it needs about 14 alarms a year to catch half of them. Onset is where count regression helps most: for S-onset, ridge on log counts reaches 0.89 against 0.26 for persistence, but this rests on 5 positive weeks. For A-onset no model is clearly useful (interval 0.04–0.80). At $h = 2$ and $h = 4$ persistence weakens (A: 0.89, 0.77) and the best models stay close to it (Appendix figures).

Negative results. Weather alone is weak: the weather-only logistic model scores 0.34 (A), 0.16 (S) and 0.13 (G), close to the base rates of 0.14, 0.14 and 0.12. Adding weather to case history does not improve on cases alone at $h = 1$ (A: 0.89 vs 0.92). Trees are not better than the linear models (CART 0.77 and random forest 0.84 for A). The lasso keeps some lagged humidity terms, but given the weak humidity cross-check (Section 2) we do not read this as a weather effect. The count models' one-week-ahead error is MAE 33 cases/week (RMSE 60) for ridge on log counts, against a development mean of 242 cases/week; at $h = 4$ the MAE is 66.

Final period, 2022. 2022 was Singapore's second-largest year in the series, with 32,130 cases. The models chosen on development keep their ranking: persistence and the selected model score 0.97–0.98 for A and S, and the growth model scores 0.31 against 0.14 for persistence (8 positives). The onset results rest on 2 positives each and cannot separate models. **Failure case.** For A-onset, the Poisson model raised alarms at the June–July 2022 peak. From the origin 2022-07-04 it predicted 1,383 cases

($P = 0.52$), but cases fell to 918. The model extrapolated growth into a turning point. It also missed the onset week of 2022-01-17 ($P = 0.18$, just below the alarm threshold).

Robustness. With a one-week reporting delay, PR-AUC for A falls from 0.93 to 0.89 (persistence) and from 0.93 to 0.81 (Poisson GLM); S changes little (0.97 to 0.94). On the brief's 2014–2018 file (test years 2017–2018), A has 7 positives (logistic 0.80 vs persistence 0.79), growth has 21 (0.29 vs 0.22), and the seasonal targets have *no* positives. This file alone cannot evaluate a surge warning, which is why it serves as a cross-check rather than the main analysis.

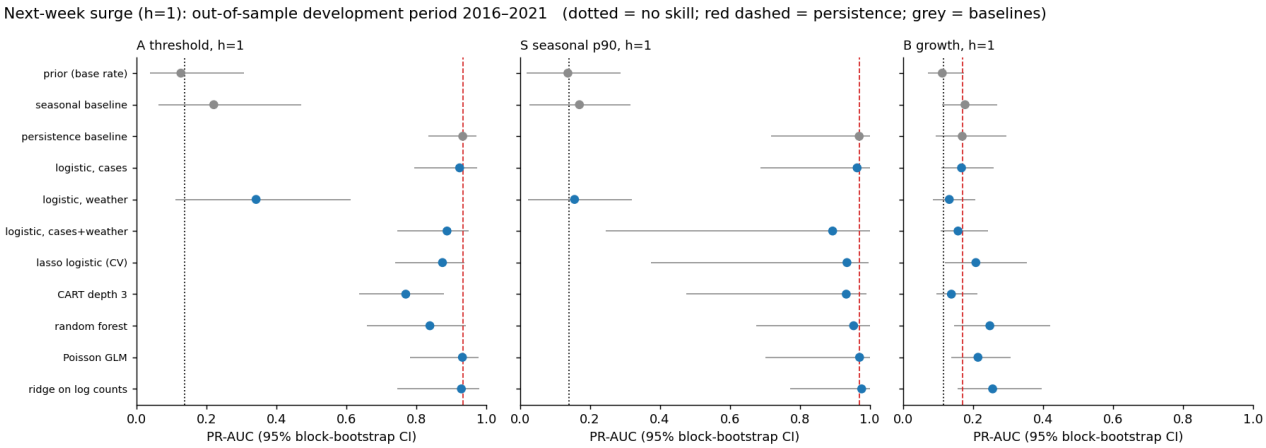


Figure 2: Development-period PR-AUC with 95% block-bootstrap intervals, $h = 1$. Dotted line: no-skill base rate; red dashed line: persistence baseline.

5 Discussion

Answer to the question. Recent case counts can flag weeks that will exceed a level threshold next week with high precision and recall. However, most of this skill comes from persistence: a simple rule "we are already above, or close to, the threshold" is as good as any model. This is useful for situational awareness but is not early warning. Where early warning is actually needed (growth and onset surges), skill is modest (growth PR-AUC about 0.26 against a base rate of 0.12) or rests on very few events (S-onset, 5 positives). Weather from one station adds no measurable value at 1–4 weeks once recent counts are known, which is consistent with weather acting on dengue through longer lags that are already reflected in the case trend.

Defence points from the brief. *Reporting revisions:* two MOH releases differ by at most 8 cases/week, and a simulated one-week delay reduces PR-AUC only moderately; revisions are therefore not the main risk. *Immunity and serotypes:* large epidemics such as 2020 and 2022 depend on population immunity and circulating serotypes, which are not in the data; a model trained on past epidemics can mistake the timing of a peak, as in the July 2022 failure case. *Few outbreaks:* onset targets have 5–12 development positives, so their intervals are wide and one week can change the ranking; we report them as indicative only. *Cost of false alarms:* at the 0.20 threshold (miss cost = 4 false alarms) the level targets cost about 10–11 alarms a year and growth about 14; the trade-off for other cost ratios is in `alarm_cost_tradeoff.csv`.

Limitations. National counts hide local clusters; notified cases depend on testing and health-seeking behaviour (including the COVID-19 period); there is one weather station and the humidity series is reanalysis data with only moderate agreement with NEA. The 2022 period was seen in exploratory runs and the rolling evaluation trains on earlier 2022 outcomes, so it is a final-period check, not an independent test. All statements are predictive, not causal.

Ethics. All inputs are aggregated public data, so privacy risk is low. The main risks are operational: false alarms divert inspection and communication resources, while misses delay response. The 4:1 cost ratio is our assumption, not elicited from decision-makers. The scores should support, not replace,

human review by public-health staff, and any deployment would need prospective validation and monitoring for changes in reporting.

6 Conclusion

A transparent persistence rule already flags most next-week level surges in Singapore, and lagged weather does not improve on recent counts. Statistical learning adds value mainly for growth and onset surges, where count regression on log cases beats the baselines, but these gains rest on few events and wide intervals. A useful next step is prospective evaluation with post-2022 counts, spatial cluster data and serotype information.

7 Reproducibility and contributions

Code, configuration, processed weekly snapshots, figures and data provenance are provided with this report. The main analysis is reproduced with `cd pipeline && python run.py all` (about 9 hours with weekly refits); `python run.py evaluate` re-scores the saved out-of-sample predictions in minutes. The 2014–2018 analysis uses `-config config_brief.yaml`. Random seeds are fixed. Raw files are downloaded by the pipeline rather than redistributed; retrieval dates, hashes and licences are recorded in `pipeline/data/raw/MANIFEST.json`, and the data dictionary is in `docs/DATA.md`.

Contributions. [To be completed by the group: verified division of work for Tran Duy Tai, Do Nhat Thanh and Nguyen Thi Quynh Thu.]

AI-use disclosure. An AI coding assistant (Copilot SDK agent in VS Code; earlier sessions with Claude Code) was used to write and debug pipeline code, convert the course PDFs to text, and draft documentation, report text and slides. The group chose the research question, surge definitions, models and validation design. All reported numbers are read from pipeline output files, and the references were checked against their DOIs. Details, representative prompts and the checks performed are in `AI_USE.md`.

References

Yien Ling Hii, Huaiping Zhu, Nawi Ng, Lee Ching Ng, and Joacim Rocklöv. Forecast of dengue incidence using temperature and rainfall. *PLoS Neglected Tropical Diseases*, 6(11):e1908, 2012. doi: 10.1371/journal.pntd.0001908.

Meteorological Service Singapore. Historical daily records. National Environment Agency, station S24 Changi. URL `http://www.weather.gov.sg/climate-historical-daily/`. Accessed 2026-11-06.

Ministry of Health, Singapore. Weekly number of dengue and dengue haemorrhagic fever cases. data.gov.sg dataset d_ac1eecf0886ff0bceefbc51556247015, 2024a. URL `https://data.gov.sg/datasets/d_ac1eecf0886ff0bceefbc51556247015/view`. Accessed 2026-11-06.

Ministry of Health, Singapore. Weekly infectious disease bulletin cases. data.gov.sg dataset d_ca168b2cb763640d72c4600a68f9909e, 2024b. URL `https://data.gov.sg/datasets/d_ca168b2cb763640d72c4600a68f9909e/view`. Accessed 2026-11-06.

Open-Meteo. Historical weather api. URL `https://archive-api.open-meteo.com/v1/archive`. CC BY 4.0; accessed 2026-11-06.

Yuan Shi, Xu Liu, Suet-Yheng Kok, Jayanthi Rajarethinam, Shaohong Liang, Grace Yap, Chee-Seng Chong, Kim-Sung Lee, Sharon S. Y. Tan, Christopher Kuan Yew Chin, Andrew Lo, Waiming Kong, Lee Ching Ng, and Alex R. Cook. Three-month real-time dengue forecast models: An early warning system for outbreak alerts and policy decision support in Singapore. *Environmental Health Perspectives*, 124(9):1369–1375, 2016. doi: 10.1289/ehp.1509981.

A Additional figures

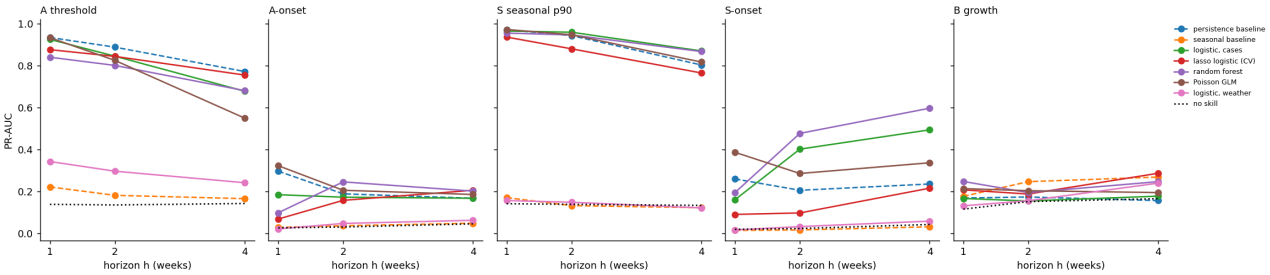


Figure 3: Development PR-AUC by horizon for each surge definition.



Figure 4: Development vs. 2022 PR-AUC for the baselines and the development-selected model, $h = 1$.

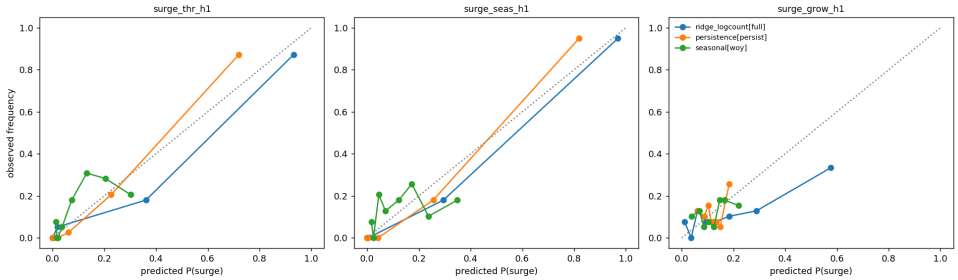


Figure 5: Calibration (development period, $h = 1$) of the best model, persistence and the seasonal baseline.